

Module 3

Processes, Pipes & Redirection

Learning objectives

- Stop foreground jobs with Ctrl+C
- Redirect stdout/stderr to log files
- Chain commands with pipes
- Search simulator logs with grep

Prerequisites

- Module 2 (permissions, executable scripts)

Learning path

Learning Path

Diagram: Learning Path

(render with mmdc when available)

flowchart LR

A[Processes] --> B[Redirection]

B --> C[Pipes & grep]

C --> D[Log analysis]

D --> E[Module 4]

Course context

- Save build/sim output: `command > run.log 2>&1`
- Filter errors: `grep -i error run.log`
- Exit status `$?` indicates success or failure

Core topics

Self-check

```
$ ./scripts/module3.sh --check
```

Terminal: module_check

```

[0;36m=====
[0;36mModule 3: Environment self-check
[0;36m=====

[0;32m[OK] ps available
[0;32m[OK] grep available
[0;32m[OK] tail available
[0;32m[OK] module3 directory exists: /home/yongfu/proj/universal-verification-methodology/
[0;32m[OK] processes/show_pid.sh exists
[0;32m[OK] redirection/print_both.sh exists
[0;32m[OK] pipes/sample.log exists
[0;32m[OK] troubleshooting/failing_cmd.sh exists
[0;32m[OK] combining/combine_demo.sh exists

[0;32m[OK] All required checks passed.
```

Capture full log

```
./scripts/run_tests.sh > logs/run.log 2>&1  
tail -f logs/run.log
```


Searching logs

```
$ cd module3/examples/pipes && grep ERROR sample.log
```

Terminal: grep_errors

```
[ERROR] Assertion failed at line 42: expected 0x0, got 0x1  
[ERROR] Mismatch in output file  
[ERROR] 2 tests failed, 1 passed
```

Redirection demo

```
$ cd module3/examples/redirection && chmod u+x print_both.sh &&  
  ./print_both.sh > /tmp/m3_out.log 2> /tmp/m3_err.log && head -2  
  /tmp/m3_out.log
```

Terminal: redirection

This line goes to stdout (standard output)
This line also goes to stdout

Practice

Exercise scaffold

```
./scripts/module3.sh --scaffold
```

Exercises

- `grep ERROR` on `~/unix_practice/sample.log`
- Try `2>&1 | tee debug.log`
- Check `echo $?` after a command

Summary & next steps

- Processes and exit status
- Redirection and pipes
- Next: Module 4 — shell scripting
- Next: Module 4: Shell Productivity & Basic Scripting